

Juan Carlos Dominguez - Resume

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Work Experience

Electronics Technician I | Arcata Associates - NASA Armstrong Flight Research Center September 2025 - Present

- Enabled the successful execution of 50+ NASA flight research missions by operating, maintaining, and optimizing telemetry systems, mission control infrastructure, and enterprise data distribution systems, sustaining 99.9% operational availability across real-time and post-flight mission phases.
- Actively contributed to the end-to-end mission data pipeline, interfacing with real-time telemetry streams, post-flight recording systems, and mission databases to ensure 99.9% data accuracy, completeness, and traceability across terabytes of high-volume flight test data, directly supporting safety, performance, and engineering analysis for 50+ research missions.
- Performed automated data validation, quality assurance, and anomaly detection on telemetry datasets using Python and command-line tools, reducing data integrity issues by 45%, detecting errors pre-analysis, and preventing downstream reprocessing delays for engineering teams.
- Compiled, configured, and integrated real-time C-based telemetry processing and display systems using Linux and Windows command-line environments, improving system performance stability by 30% and enabling rapid fault isolation and resolution during live mission operations.
- Resolved complex computer, network, switch, and router issues supporting mission operations, restoring critical services in under 20 minutes for 95% of incidents, minimizing downtime in real-time mission environments.

Computer Operator | ASRC Federal - NASA Armstrong Flight Research Center March 2025 - September 2025

- Orchestrated the operation, engineering, and optimization of Telemetry and Radar Acquisition Processing Systems, Mission Control Centers, and mobile operations facilities, sustaining 95% system uptime across 50+ NASA Flight Research Missions.
- Enhanced and streamlined high-volume mission databases for telemetry and onboard data, cutting retrieval times by 40% and accelerating engineering analysis speed by 45% for critical, time-sensitive mission assessments.
- Managed post-flight data processing pipelines for all real-time telemetry and aircraft onboard datasets, achieving zero data loss, meeting 100% of NASA's delivery deadlines, and reducing processing cycle times by 40% through workflow automation.
- Performed acceptance testing, diagnostics, and repairs on data acquisition system components, ensuring compliance with specifications and reducing equipment-related mission delays by 25% through proactive fault detection and resolution.
- Conducted pre- and post-flight system checks for instrumentation and data acquisition systems, identifying and correcting issues ahead of schedule to maintain 100% on-time mission readiness for flight test operations.

Projects

Real-Time International Space Station Tracking Tool | HTML, CSS, JavaScript, Open Notify API Summer 2025

- Crafted and executed a real-time ISS tracking tool, integrating Open Notify API to dynamically visualize the station's trajectory with 25% improved positional accuracy, enhancing user engagement and data reliability.
- Optimized and automated API data retrieval and continuous updates, ensuring 99% uptime while streamlining response efficiency, reducing latency, and enabling seamless real-time tracking without manual intervention.

Data-Driven Insights for Early Breast Cancer Diagnosis | Python, SQL, Tableau Winter 2024

- Architected end-to-end SQL pipelines to efficiently process & aggregate large-scale diagnostic datasets, achieving a 40% reduction in data preparation time while ensuring data accuracy and scalability. Leveraged Python for advanced statistical modeling and in-depth feature correlation analysis, enabling precise identification of key diagnostic predictors.
- Uncovered transformative insights by analyzing diagnostic trends and patterns, leading to a 25% improvement in malignancy risk assessment accuracy. This work significantly contributes to refining diagnostic protocols and empowering clinicians to make more informed, data-backed decisions, improving patient care outcomes.

Technical Skills

Programming Languages: HTML, CSS, JavaScript, Python, SQL, R, XML

Technologies/Frameworks: Pandas, NumPy, TensorFlow, Flask, React, Node.js

Systems & Data: Linux, Windows CLI, Telemetry Data Pipelines, REST APIs

Education

- M.S. Computer Science, *Southern Methodist University*

Graduated